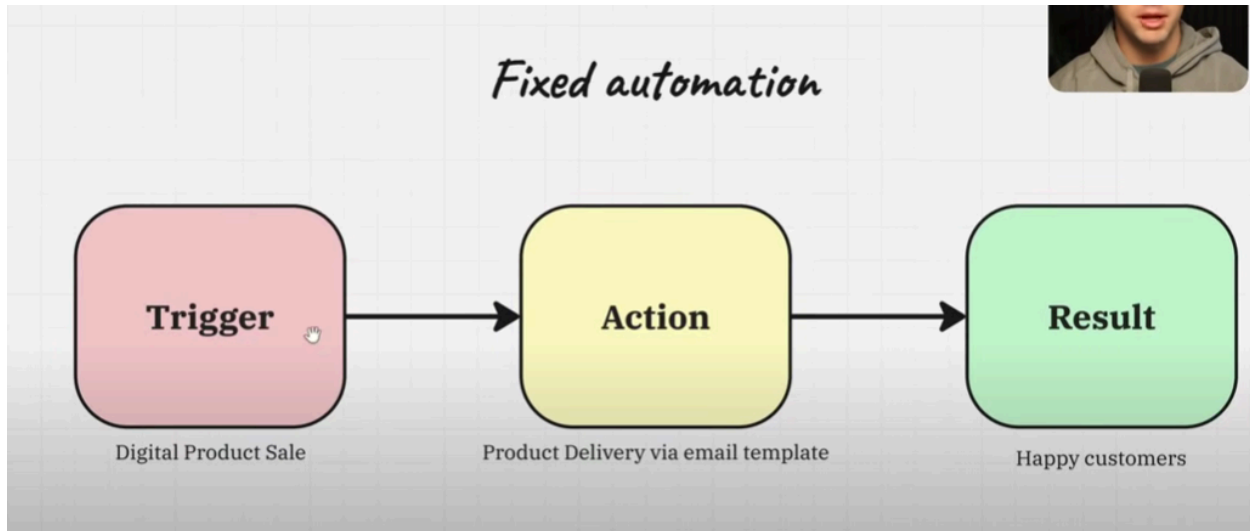


Section 6 - Agent and Automation Concept



Key Concepts Covered

Automation vs Agent

Feature	Automation	Agent
Nature	Predefined	Adaptive
Instruction Level	Exact	Goal-driven
Decision Making	Fixed	Dynamic
Flexibility	Low	High
Use Case	Repetitive tasks	Personalized, dynamic scenarios
Example	Send product email after Stripe payment	Send personalized email based on past purchases and market trends

Practical Examples

1. Fixed Automation Example

- Trigger: Stripe purchase
- Action: Send fixed email with product link
- Result: Simple, high-ROI automation for product delivery
- Uses static fields: customer name, email, product name

2. Agentic Workflow Example

- Trigger: Stripe purchase
 - Action: AI writes personalized email based on:
 - Purchase history
 - Company backend data
 - Customer timeline
 - Market trends
 - Result: Upsells, dynamic content, more sales, higher engagement
-

Agent Capabilities

- Can use tools and external/internal data (e.g. Airtable, calendar, Slack, Google Sheets)
- Makes decisions aligned with user's **goals**
- Learns and reasons over time
- Acts as a **personal assistant**

▼ Agent Use Case Built in Tutorial

Objective: Build an AI agent that reads emails, labels them, and drafts replies based on your goals and calendar

Functional Requirements:

- Read incoming Gmail messages

- Categorize & label emails
- Draft responses based on:
 - User's goals (stored in Airtable)
 - User's calendar events
- Responses saved as drafts (not auto-sent)

Instruction Basics

There are a couple of things to keep in mind here

Basic instruction framework:

1. **Role setting** - 'You are an agent that specializes in [x,y,z].
2. **Goals**- your main goal is to create [x,y,z].
3. **Guidelines** -Don't do [x,y,z]. Ensure responses follow [x,y,z],
4. Input and output example (**in json if needing structure**)

JSON basics

JSON stands for javascript object notation, and it's a lightweight format for storing and transporting data.

With JSON, AI can parse and structure your data into multiple fields you easily map.

This is great for filtering, routing, exporting fields and more.

Think of it like a form

You don't tell people to input all of their information into a single form field.

You break it up into 'key-value' pairs.

Structure Output parser

Here, we can tell the agent which 'field or 'key value' pair we want.

This is crucial for consistency and for ensuring we have multiple, unique fields we can use for mapping to other applications.

In this case we need a 'label' and a response to be in separate fields.

Steps to Build the Agent

1. Setup Gmail Trigger in n8n

- Use trigger: `Gmail → On Message Received`
- Pulls: sender, body, subject, date
- Optional: Add filter (e.g., only unread emails)

2. Fetch Full Email Details

- Use: `Gmail → Get Message`
- Extract rich data: HTML body, headers, full content

3. Prepare AI Agent Node

- Add AI Agent node in n8n canvas
 - Choose a model provider (e.g., OpenAI, Groq, Mistral, etc.)
 - Use OpenAI's chat model with your API key
-

Connecting OpenAI API in n8n

1. Go to: <https://platform.openai.com>
 2. Sign in or create an account
 3. Go to **Settings → API Keys**
 4. Add billing (credits required)
 5. Create new secret key
 6. Copy & paste into n8n credentials
 7. Test and confirm connection
-

What the Agent Can Eventually Do

- Read incoming email like:
"Hey Drake, can we link up tomorrow at 10?"
- Reference goals & calendar:
 - E.g., deny meeting due to conflict
- Draft customized responses using past history + real-time data
- Apply labels/categories based on content

▼ 🚀 Part 2: Building the AI Agent in n8n – Email Labeling & Reply Drafting Agent

🔧 Step-by-Step Breakdown

✅ 1. Set Up Gmail Trigger

- **Node:** Gmail → "On New Message"
- **What it does:** Detects new email in inbox
- **Output:** Basic email data – subject, snippet, sender email

🔄 2. Fetch Full Email Content

- **Node:** Gmail → "Get Message"
- **Input:** Email ID from previous step
- **Output:** Full content including plain text, HTML, full headers

💡 Why Get Full Message?

- Needed to feed detailed context to the AI agent
- Enables better classification & personalized reply generation

✨ 3. Add AI Agent Node

- **Node type:** AI Agent (from n8n)
- **Model used:** OpenAI GPT-4 (or other chat-compatible model)

- **Credential:** Your OpenAI API key
-

What Should the Agent Do?

Agent's Goals:

- Read and understand the email
 - Categorize (e.g., "Meeting Request", "Spam", "Urgent")
 - Draft a reply **aligned with your goals and calendar**
-

4. Connect Agent with Real Data

You'll provide the AI agent with 3 key contexts:

- The **email message content**
 - Your **personal goals** (from Airtable or static list)
 - Your **calendar events** (from Google Calendar node)
-

5. Prompt Design for Agent

Prompt the agent with:

```
text
CopyEdit
You are my email assistant. You will:
1. Categorize this email.
2. Suggest a label from: ["Important", "Ignore", "Meeting Request", "Other"]
3. Draft a reply based on:
  - My current goals
  - My calendar
  - My past conversations (if available)

GOALS:
{{YourGoals}}
```

CALENDAR:
{{YourCalendarEvents}}

EMAIL:
From: {{email.from}}
Subject: {{email.subject}}
Body: {{email.body}}

Example Output from Agent:

- **Category:** Meeting Request
- **Suggested Label:** Important
- **Draft Reply:** "Hey John, thanks for reaching out! Tomorrow at 10 doesn't work for me. I'm available Thursday at 11. Let me know if that works!"



6. Test the Agent

- Trigger a test email
- Watch output:
 - Email label
 - AI-generated draft response
- Optional: Push response as a **Gmail draft**



Tools You Can Use Along with the Agent

- **Airtable:** Store goals, clients, tags
- **Google Calendar:** Check availability
- **Vector store (Weaviate):** To look up past emails or messages
- **Webhook or API nodes:** To trigger replies, Slack notifications, CRM updates, etc.



Cleaning Output (Optional)

- Use `Code Node` to clean up output JSON
- Format label, remove line breaks, handle AI errors

Agentic AI Email Agent Using n8n – Step-by-Step Summary

Overview

You are building an **AI agent using n8n** that can:

- Understand user input (intent)
 - Act based on goals (autonomously or via feedback loop)
 - Perform actions (like sending emails)
 - Learn from the results
-

Tools Used

- **n8n** (open-source automation tool)
 - **OpenAI** (for intent classification and decision-making)
 - **Airtable** (to store memory/state if needed)
 - **Email node** (to send actual emails)
-

Agent Workflow Components

1. Trigger

- Use a **Webhook trigger** or a **Manual trigger** for testing.

2. User Input / Goal

- User provides a message like:

"Send an outreach email to 5 clients from the coffee industry about our herbal extract product."

3. OpenAI - Intent Extraction

- Ask OpenAI to:
 - Extract goal (send email)
 - Extract key details (industry, target, message)
- Use structured prompt to get a JSON output.

Prompt Example:

"You are a helpful assistant. Analyze the user request and extract:

- Action
- Audience
- Message content

Return in JSON format."

4. Memory / Agent Loop (Optional)

- Use **Airtable** or **n8n Variables** to store steps taken
- Let the agent plan next steps like:
 - Write email draft
 - Approve content
 - Send to target audience

5. Generate Email Content

- Pass details to OpenAI to generate an email draft.

Prompt Example:

"Write a professional marketing email for [herbal extract product] targeting coffee shops. Highlight health benefits and offer a free sample."

6. Send Email

- Use **Email node** or **Gmail node**
- Input:
 - Subject
 - To address (can be dynamic)
 - HTML/plain text content

Agent Intelligence Tip

The agent can:

- Make decisions ("should I send the email now?")
- Ask for feedback if unsure
- Loop until the task is complete or confirmed

Agent Capabilities Recap

Functionality	Agent Behavior	
Categorization	Labels the email dynamically	
Reasoning	Considers past responses, goals, and schedule	
Output Handling	Returns reply + label in clean structure	
Customizable Prompt	Easy to swap calendar, tone, intent	